

The Fourier-Multiplier Induction Map Is Not Well Defined in General

Full solution packet for arXiv:2011.00146

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Status: full solution likely valid. For every uniform lattice $\Gamma < \mathrm{SL}(3, \mathbb{R})$ covered by the source theorem, some $m \in MA(\Gamma)$ has induced function $\widehat{m} \notin MA(\mathrm{SL}(3, \mathbb{R}))$.

1 Source question and context

Battseren considers a lattice Γ in a locally compact group G , a normalized Borel fundamental domain Ω , and the induction formula

$$\Phi : \ell^\infty(\Gamma) \longrightarrow C_b(G), \quad \Phi(m)(x) = \widehat{m}(x) = \int_{\Omega} m(\gamma(x\omega)) d\omega. \quad (1)$$

Here $\gamma : G \rightarrow \Gamma$ is determined by the decomposition $G = \bigsqcup_{y \in \Gamma} \Omega y$. Lemma 18 of the arXiv version proves that (1) is well defined and that it restricts contractively to the Fourier algebra and the completely bounded multiplier algebra [1]. For a uniform lattice Γ in $G = \mathrm{SL}(3, \mathbb{R})$, Theorem A proves that it does not give a bounded map $MA(\Gamma) \rightarrow MA(G)$.

Immediately after that proof, the source asks in Remark 19:

“We do not know if the map $\Phi : MA(\Gamma) \rightarrow MA(G)$ is well defined.”

The 2024 journal version retains the same question as Remark 4.2 and labels the unboundedness statement Theorem B.

Thus, if $L(x) \leq n - 2c$, we have $\dot{\varphi}_n(x) = 1$. Therefore, the left hand side of (4) grows exponentially while the right hand side is polynomial, which gives the desired contradiction. $\square$

Remark 19. *We do not know if the map $\Phi : MA(\Gamma) \rightarrow MA(G)$ is well defined.*

Theorem B. *Let Γ be a uniform lattice in $G = \mathrm{SL}(3, \mathbb{R})$. Then $M_0A(\Gamma)$ is a proper subalgebra of $MA(\Gamma)$.*

Figure 1: Remark 19 in the original arXiv paper, PDF page 10.

2 Proof intuition

The source already supplies the hard harmonic analysis: boundedness of a hypothetical restriction is impossible. The missing step is purely functional analytic. Formula (1) is continuous in the ambient

sup norms, while each Fourier multiplier norm is stronger than the sup norm. Therefore, if the restriction were defined on every multiplier, any multiplier-norm convergent graph sequence would have the same uniform limit on both sides. Its graph would be closed, and the closed graph theorem would force boundedness—exactly what the source theorem excludes.

3 The closed-graph lemma

Lemma 1. *For every locally compact group H and every $m \in MA(H)$,*

$$\|m\|_\infty \leq \|m\|_{MA(H)}.$$

Proof. Fix $x \in H$. Choose a unit vector $\xi \in L^2(H)$ and define the coefficient function

$$u(y) = \langle \lambda(y)\xi, \lambda(x)\xi \rangle.$$

Then $u \in A(H)$, $u(x) = 1$, and $\|u\|_{A(H)} \leq 1$. Since the Fourier algebra norm dominates the sup norm,

$$|m(x)| = |(mu)(x)| \leq \|mu\|_{A(H)} \leq \|m\|_{MA(H)} \|u\|_{A(H)} \leq \|m\|_{MA(H)}.$$

Take the supremum over x . □

Lemma 2. *Let $\Gamma < G$ be a lattice and let Φ be given by (1). If*

$$\Phi(MA(\Gamma)) \subseteq MA(G),$$

then the restriction $\Phi : MA(\Gamma) \rightarrow MA(G)$ is bounded.

Proof. The normalized integral formula gives the ambient contraction

$$\|\Phi(m)\|_\infty \leq \|m\|_\infty \quad (m \in \ell^\infty(\Gamma)). \quad (2)$$

Suppose $m_n \rightarrow m$ in $MA(\Gamma)$ and $\Phi(m_n) \rightarrow h$ in $MA(G)$. By Lemma 1, both convergences imply uniform convergence. Equation (2) also gives

$$\Phi(m_n) \longrightarrow \Phi(m) \quad \text{uniformly on } G.$$

Uniqueness of uniform limits yields $h = \Phi(m)$. Hence the graph of the restriction is closed. Both multiplier spaces are Banach spaces, so the closed graph theorem [2, Theorem 2.15] makes the restriction bounded. □

4 Full negative answer

Theorem 1. *Let Γ be a uniform lattice in $G = \mathrm{SL}(3, \mathbb{R})$ and let Φ be the induction formula used in [1]. Then*

$$\Phi(MA(\Gamma)) \not\subseteq MA(G).$$

Equivalently, there exists $m \in MA(\Gamma)$ for which the induced bounded continuous function $\hat{m} = \Phi(m)$ is not a Fourier multiplier of G .

Proof. Assume instead that $\Phi(MA(\Gamma)) \subseteq MA(G)$. By Lemma 2, the restriction $\Phi : MA(\Gamma) \rightarrow MA(G)$ is bounded. This contradicts Theorem A of the arXiv source (Theorem B of the published version), which proves that the same restriction cannot be bounded for such a uniform lattice. Therefore the restriction is not everywhere well defined. □

5 Verification, novelty, and limitations

The proof has no numerical or finite-search dependency. Its four interfaces are explicit: the source integral formula gives (2); Lemma 1 gives both continuous embeddings into sup norm; uniqueness of uniform limits closes the graph; and the source theorem supplies the unboundedness contradiction. The published proof itself begins by assuming that the restriction is “well defined and bounded,” so the quantifiers match the contradiction used here.

A bounded novelty search on 22 July 2026 covered the run’s four cheap indexes; the exact arXiv id, title, author, and well-definedness phrase; arXiv searches combining $MA(\Gamma)$, $MA(G)$, induction, well defined, and closed graph; the open-access 2024 journal version; and the author’s later arXiv:2408.09638 on M_d multipliers and lattices. No later resolution or occurrence of this closed-graph argument was found.

Novelty confidence is moderate because the argument is elementary and may be folklore or implicit in the published theorem’s wording. The explicit Remark 4.2 nevertheless remains in the published paper. The theorem here is limited to the induction formula and the uniform $SL(3, \mathbb{R})$ lattices covered by the source unboundedness theorem; it does not classify other lattice pairs and does not resolve the paper’s amenability conjecture.

Human review should focus on the interpretation and quantifiers of the source unboundedness theorem. Once those are accepted, the closed-graph step is complete.

References

- [1] B.-O. Battseren, *On the growth of Fourier multipliers*, arXiv:2011.00146; J. Noncommut. Geom. 18 (2024), no. 4, 1209–1228, <https://doi.org/10.4171/JNCG/551>.
- [2] W. Rudin, *Functional Analysis*, second edition, McGraw–Hill, 1991.

Source evidence

4. PROOF OF THEOREM A AND THEOREM B

As a motivation to investigate tame cuts, we provide two applications of tame cuts. The first one is about to continuity of the induction map. The following lemma shows the importance of induction map and is used to prove that amenability, a-T-menability, weak-amenability, and Kazhdan's property (T) are inherited by lattices.

Lemma 18 (Haagerup, [Haa16]). *Let Γ be a lattice in a locally compact group G . Let $\sigma : G/\Gamma \rightarrow G$ be a Borel cross section, and $\Omega \subseteq G$ its image. Denote by μ_Γ the counting measure on G corresponding to Γ . Then*

$$\Phi : \ell^\infty(\Gamma) \rightarrow C_b(G), \quad \varphi \mapsto \hat{\varphi} = \mathbb{1}_\Omega * (\varphi \mu_\Gamma) * \tilde{\mathbb{1}}_\Omega$$

is well defined. Moreover, the restrictions of this map give bounded linear maps

$$\Phi : A(\Gamma) \rightarrow A(G) \quad \text{and} \quad \Phi : M_0 A(\Gamma) \rightarrow M_0 A(G).$$

Moreover, if φ is a positive definite function on Γ , so is $\hat{\varphi}$ on G .

As we have a disjoint union $G = \bigsqcup_{y \in \Gamma} \Omega y$, the map $\gamma : G \rightarrow \Gamma$ such that $x \in \Omega \gamma(x)$ is well defined, and we have

$$\Phi(x) = \int_\Omega \varphi(\gamma(x\omega)) d\omega \quad \text{for all } x \in G.$$

where $d\omega$ is the normalized Haar measure on G so that $\int_\Omega d\omega = 1$.

Theorem A. *Let Γ be a uniform lattice in $G = SL(3, \mathbb{R})$. Then the map*

$$\Phi : MA(\Gamma) \rightarrow MA(G), \quad \varphi \mapsto \hat{\varphi} = \mathbb{1}_\Omega * (\varphi \mu_\Gamma) * \tilde{\mathbb{1}}_\Omega$$

is not bounded.

Proof. Recall that the length functions ℓ_S , $\ell_C|_\Gamma$, and $L|_\Gamma$ are all equivalent on Γ (cf. Section 2.3). We choose L as the main length function on both Γ and G .

By contradiction, suppose the map $\Phi : MA(\Gamma) \rightarrow MA(G)$ is bounded. By Theorem 9 Γ has RD, and a fortiori characteristic tame cuts, so there is finitely supported characteristic functions φ_n such that

Figure 2: The induction formula and source Theorem A, arXiv PDF page 9.